EpoAdapt: Adaptive Evolutionary Pareto Optimization with Asymmetric Feasibility Control

Anonymous Authors[1]

Abstract

Subset selection with a monotone submodular objective under a linear cost constraint is a fundamental problem with applications in sensor placement, document summarization, sparse regression, maximum coverage, and influence maximization. While recent evolutionary approaches such as EPOL provide strong theoretical guarantees, they rely on uniform $1/n$ exploration and expensive Monte Carlo–based objective estimation, which can limit effectiveness under a fixed evaluation limit. We propose EPOADAPT, an adaptive evolutionary Pareto optimization algorithm that introduces statistically grounded asymmetric feasibility-aware mutation, together with submodularly informed Pareto initialization and adaptive control of the number of parallel subproblems K_B, to concentrate search on promising feasible regions. For influence maximization, EPOADAPT employs a hybrid strategy that combines theoretically guaranteed reverse-reachable set screening with lightweight Monte Carlo refinement, substantially reducing objective evaluations while preserving the approximation guarantees of existing evolutionary Pareto methods as a special case. Experiments on standard influence maximization benchmarks, using the same datasets and evaluation protocols as prior work, demonstrate that EPOADAPT consistently achieves higher solution quality than prior methods under the same evaluation limit.

1. Introduction

Subset selection under a budget constraint is a fundamental problem in which we must allocate limited resources to a subset of candidates while maximizing a global utility function. In its general form, this NP-hard problem arises across diverse cost-aware resource allocation tasks, including maximum coverage (Feige, 1998), influence maximization (Kempe et al., 2003), sparse regression (Tropp, 2004), sensor placement (Krause et al., 2008), document summarization (Lin & Bilmes, 2011), budget-constrained advertising (Ashkan et al., 2015), and human-assisted learning (Liu et al., 2023). We focus on the influence maximization (IM) problem under the Independent Cascade (IC) model, where V denotes the set of candidate seed nodes in a directed social graph and $\sigma(X) := \mathbb{E}[|IC(X)|]$ denotes the expected number of activated nodes when seeding $X \subseteq V$. The monotone submodular structure of $\sigma(\cdot)$, together with established benchmarks from recent evolutionary methods such as EPOL (Liu & Qian, 2025), makes IM under IC particularly suitable for controlled comparisons, allowing us to evaluate EPOADAPT against a broad range of algorithms using identical datasets, budget settings, and evaluation protocols.

For this problem, algorithms with theoretical approximation guarantees fall into several families. Greedy methods remain the primary approach: the Generalized Greedy Algorithm repeatedly selects the item maximizing marginal gain per unit cost and returns the better of the final greedy solution and the best singleton. Depending on the specific variant and analysis, this yields constant-factor approximations, with guarantees ranging up to $1 - 1/\sqrt{e} \approx 0.39$ in the strongest cases (Krause & Guestrin, 2005; Lin & Bilmes, 2010; Zhang & Vorobeychik, 2016). Greedy+ strengthens this strategy by considering single-item expansions of intermediate solutions, improving both empirical performance and worst-case bounds (Yaroslavtsev et al., 2020), while 1-guess-Greedy+ adds a limited partial-enumeration step to achieve a 0.6174 approximation (Feldman et al., 2023). Although polynomial-time algorithms achieving a $1 - 1/e \approx 0.632$ worst-case approximation ratio exist, their computational requirements remain impractical for real-world applications (Sviridenko, 2004; Badanidiyuru & Vondrák, 2014; Ene & Nguyen, 2019).

This gap between theory and scalability has motivated the application of evolutionary algorithms (EAs) to budgeted submodular maximization. Rather than committing to a

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

For ICML 2026 reviewers: When reviewing this paper, please follow the policy for the use of LLMs as specified on your Reviewer Console. Note that the assigned LLM policy might differ from the policy you indicated during reviewer registration.

single greedy trajectory, Pareto-based EAs use randomized variation and population-based selection to explore a broader set of cost–value tradeoffs, and several recent EA designs come with provable approximation guarantees. Pareto-based approaches such as POMC maintain a population of non-dominated solutions trading off objective value and cost, achieving a $(1/2)(1-1/e) \approx 0.316$ approximation (Qian et al., 2017). FPOMC introduced greedy selection into this framework while retaining the same approximation ratio (Bian et al., 2021), and single-objective designs EAMC and EVO-SMC optimized surrogate combinations of objective and cost, achieving $(1/2)(1-1/e)$ and 0.5 approximations respectively (Bian et al., 2020; Zhu et al., 2024). EPOL represents a significant theoretical and empirical advance: by decomposing the problem into a parallel family of anchored residual subproblems, it achieves a 0.6174 approximation guarantee and sets the empirical state of the art on maximum coverage and IM benchmarks under its evaluation protocol (Liu & Qian, 2025).

Despite these advances, evolutionary methods face a fundamental challenge in the evaluation-limited regime. For IM, Monte Carlo estimation of $\sigma(\cdot)$ is expensive, making objective evaluations the primary bottleneck. Uniform $1/n$ mutation generates infeasible offspring near the constraint boundary, and pure Monte Carlo estimation consumes evaluations even when approximate estimates would suffice for search guidance. These inefficiencies motivate adaptive mechanisms that explicitly target finite computational limits.

We introduce EPOADAPT, an adaptive evolutionary Pareto optimizer that employs Reverse Influence Sampling (RIS) for initial screening followed by lightweight Monte Carlo refinement, submodularity-guided control of the anchor count K_B, and feasibility-aware asymmetric mutation. Since EPOL emerges as a special case under the corresponding specialization, its approximation guarantees apply. On standard EPOL benchmarks, EPOADAPT consistently outperforms prior methods under stricter computational limits.

Our contributions are as follows:

- We propose submodularity-guided anchor selection that adaptively sets K_B using pairwise distinctiveness and overlap penalization.
- We design statistically grounded asymmetric mutation that promotes feasibility near the constraint boundary.
- For IM, we develop a hybrid evaluation strategy combining RIS screening with lightweight Monte Carlo refinement.
- We prove that EPOL is recovered as a special case of EPOADAPT under the corresponding specialization.
- On standard EPOL benchmarks and protocol, EPOADAPT achieves superior objective values under substantially fewer evaluations per anchored subproblem.

2. Method

Let $V = \{v_1, v_2, \ldots, v_n\}$ denote the ground set, and let $f : 2^V \to \mathbb{R}$ be a set function. The function f is *monotone* if for all $X \subseteq Y \subseteq V$,

$$f(X) \leq f(Y).$$

The function f is *submodular* if for all $X \subseteq Y \subseteq V$ and all $v \in V \setminus Y$,

$$f(X \cup \{v\}) - f(X) \geq f(Y \cup \{v\}) - f(Y).$$

We formally consider the problem of maximizing such a function subject to a linear budget constraint:

Definition 2.1. Let V denote a ground set. Given a monotone submodular function $f : 2^V \to \mathbb{R}$, element costs $c : V \to \mathbb{R}^+$, and a budget $B > 0$, define the induced linear cost of a set $X \subseteq V$ as

$$CS(X) = \sum_{v \in X} c(v).$$

The subset selection problem under a linear cost constraint is

$$\arg\max_{X \subseteq V} f(X) \quad \text{s.t.} \quad CS(X) \leq B.$$

We adopt a modular additive cost function CS, following the construction in EPOL (Liu & Qian, 2025). For a weighted adjacency matrix $W \in \mathbb{R}^{n \times n}$, let the number of positive outgoing edges from node i be

$$d_i = \sum_{j=1}^{n} \mathbb{I}_{\{W_{ij} > 0\}}.$$

The cost associated with selecting node i is defined as

$$c_i = 1 + (1 + |\epsilon_i|)\, d_i,$$

where $\epsilon_i \sim \mathcal{N}(0, 0.5^2)$ is drawn once per node and held fixed throughout optimization. For a selection vector $s \in \{0,1\}^n$ (the indicator of a set $X \subseteq V$), the total cost is

$$CS(s) = \sum_{i=1}^{n} s_i c_i = \langle s, c \rangle.$$

Under our cost model, $c_i \geq 1$ implies $K_B = \max\{|s| : CS(s) \leq B\} \leq B$, so the expected evaluation complexity bounds for FPOMC that scale as $\mathcal{O}(n^2 K_B)$ become $\mathcal{O}(B)$ for fixed n (Bian et al., 2021).

We use standard Pareto dominance to compare candidate solutions, defined as follows:

Confidential reviewer copy. This manuscript is under double-blind review by ICML 2026. Unauthorized sharing, redistribution, or disclosure is strictly prohibited.

Definition 2.2. Let x, x' be two candidate solutions evaluated under two objectives f_1 and f_2.

- x *weakly dominates* x', denoted $x \succeq x'$, if
$$f_1(x) \geq f_1(x') \ \wedge \ f_2(x) \geq f_2(x').$$
- x *dominates* x', denoted $x \succ x'$, if $x \succeq x'$ and at least one inequality is strict, i.e.,
$$f_1(x) > f_1(x') \ \vee \ f_2(x) > f_2(x').$$
- they are *incomparable* if neither $x \succeq x'$ nor $x' \succeq x$.

While EPOADAPT is suitable for the subset selection problem under budget constraints (Definition 2.1), in this paper we focus on its applicability to the IM (Kempe et al., 2003) problem. This choice is motivated by the availability of established datasets and evaluation metrics, enabling direct comparison with a wide range of existing algorithms.

For a directed graph $G = (V, E)$, let the edge activation probabilities be $p_{u,v} \in [0, 1]$ for each $(u, v) \in E$. Under the *IC* model, starting from a seed set $X \subseteq V$, we set $A_0 = X$. At each round $t = 0, 1, 2, \ldots$, every node $u \in A_t$ gets a single chance to activate each currently inactive out-neighbor $v \notin A_t$ independently with probability $p_{u,v}$; newly activated nodes form A_{t+1}. The process terminates when $A_{t+1} = A_t$. Let $IC(X)$ denote the (random) final active set produced by this process, and define the expected spread as $\sigma(X) := \mathbb{E}[|IC(X)|]$. The IM problem is then formalized as follows:

Definition 2.3. Given $G = (V, E)$, edge probabilities $p_{u,v}$, node costs $c : V \to \mathbb{R}^+$, and a budget $B > 0$, the (budgeted) IM problem is
$$\arg\max_{X \subseteq V} \ \sigma(X) \quad \text{s.t.} \quad \sum_{v \in X} c(v) \leq B.$$

2.1. The EPOADAPT Algorithm

EPOADAPT is an evolutionary adaptive Pareto-optimization algorithm for monotone submodular maximization under a linear cost constraint. In the context of IM, defined in 2.3, EPOADAPT begins by constructing singleton statistics using RIS. It first chooses an RR-set sample size θ from the standard RIS concentration guarantee (line 3) using the same ideas described in (Borgs et al., 2014; Tang et al., 2014; 2015), and then uses the resulting θ reverse reachable (RR) sets to estimate singleton gains over the ground set (line 4).

Rather than initializing the search from a single starting point, EPOADAPT determines an adaptive anchor count K_B (line 5) and selects a set of K_B anchors $\mathcal{I}$ by a diversity-aware greedy rule based on these singleton statistics (line 6). Given the anchor set $\mathcal{I}$, EPOADAPT decomposes the original instance into $|\mathcal{I}|$ anchored subproblems and solves them in parallel. Specifically, for each $i \in \mathcal{I}$, the algorithm invokes the anchored subroutine POMC_GR$(i, T, G, B, \text{PROBLEM})$ (lines 9–11), which returns a candidate solution x_i together with its objective value f_i and cost c_i. Finally, EPOADAPT aggregates the results by returning
$$x^\star \in \arg\max\{f(x_i) : i \in \mathcal{I},\ c(x_i) \leq B\},$$
i.e., the best feasible solution found across all anchored runs (lines 10–13).

SUBMODULAR_KB relies on coverage sets $\{C_i\}$ to determine an adaptive anchor count K_B. It first samples $S \subseteq I$ uniformly (line 4). When $U = |C_i \cup C_j| > 0$, the quantity
$$\rho_{ij} = 1 - \frac{|C_i \cap C_j|}{|C_i \cup C_j|}$$
inserted into $\mathcal{R}$ (lines 8–9) is the normalized dissimilarity between two coverage sets. Here $\rho_{ij} = 0$ exactly when $C_i = C_j$ and $\rho_{ij} = 1$ when $C_i \cap C_j = \emptyset$, so its magnitude directly tracks redundancy between candidates. Taking the median of $\mathcal{R}$ (line 10) yields a typical dissimilarity level that is stable to a minority of extreme pairs (near-duplicates or near-disjoint outliers), while flooring at $\rho_{\min}$ enforces a minimum exploration level so K_B does not collapse in highly overlapping regimes. After computing the adaptive anchor count K_B, we form the anchor set $\mathcal{I}$ using a greedy gain–overlap tradeoff based on singleton scores $\{f_i\}$ and coverage sets from 2-hop reachability $\{C_i\}$. Let $\mathcal{I}_0 = \emptyset$

Algorithm 1 EPOADAPT

1: **Input:** T, G; instance PROBLEM with ground set size n and budget B
2: **Output:** Best feasible solution $x^\star$
3: Build approximate coverage sets $\{C_i\}$ via 2-hop graph reachability.
4: Choose RIS sample size θ using theoretical guarantees.
5: Estimate singleton marginal gains $\{f_i\}$ using RR sets with sampling budget θ.
6: $K_B \leftarrow$ SUBMODULAR_KB$(n, \{f_i\}, \{C_i\}, G)$.
7: $\mathcal{I} \leftarrow$ select K_B anchors by diversity-aware greedy using $\{f_i\}$ and $\{C_i\}$.
8: $x^\star \leftarrow \mathbf{0}$.
9: Choose a task- and quota-aware number of workers.
10: **in parallel for each** $i \in \mathcal{I}$ **do**
11: $\quad (x_i, f_i, c_i) \leftarrow$ POMC_GR$(i, T, G, B, \text{PROBLEM})$
12: $\quad$ **if** $f_i > f(x^\star)$ **then** $x^\star \leftarrow x_i$
13: **end parallel for**
14: **return** $x^\star$

Confidential reviewer copy. This manuscript is under double-blind review by ICML 2026. Unauthorized sharing, redistribution, or disclosure is strictly prohibited.

Algorithm 2 SUBMODULAR_KB

Input: n; singleton scores $\{f_i\}$; coverage sets $\{C_i\}$; budget G; sample size s; floor $\rho_{\min}$
Output: K_B
Let I be items with defined scores; **if** $|I| < 2$ **return** $\max(1, \lfloor G/n \rfloor)$
Sample $S \subseteq I$ uniformly without replacement with $|S| = \min\{s, |I|\}$
$\mathcal{R} \leftarrow \emptyset$
for all $(i, j) \in S \times S$ with $i < j$ **do**
  $U \leftarrow |C_i \cup C_j|$
  if $U > 0$ then add $\rho_{ij} \leftarrow 1 - \frac{|C_i \cap C_j|}{U}$ to $\mathcal{R}$
end for
Compute pairwise distinctiveness over S and take the median, floored at $\rho_{\min}$
Scale the baseline $\lfloor G/n \rfloor$ by this value and cap by budget and n
return K_B

and for $t = 0, 1, \ldots, K_B - 1$ define the cumulative overlap with the current anchors as

$$\Omega_t(i) \;=\; \sum_{j \in \mathcal{I}_t} |C_i \cap C_j|.$$

At step t we select the next anchor by

$$i_t \;\in\; \arg\max_{i \notin \mathcal{I}_t} \Big(f_i - \lambda\, \Omega_t(i) \Big), \qquad \mathcal{I}_{t+1} \leftarrow \mathcal{I}_t \cup \{i_t\}.$$

Equivalently, one can maintain $\Omega_t(i)$ incrementally: after choosing i_t, update

$$\Omega_{t+1}(i) \;=\; \Omega_t(i) + |C_i \cap C_{i_t}| \quad \text{for all } i \notin \mathcal{I}_{t+1}.$$

Here $\lambda \geq 0$ controls the strength of redundancy penalization: $\lambda = 0$ reduces to selecting the top-K_B singletons by f_i, while larger λ favors anchors with less overlapping coverage.

Algorithm 3 instantiates our subroutine POMC_GR, which differs from standard POMC by (i) seeding the Pareto set with feasible greedy-extended solutions and (ii) using asymmetric feasibility-aware adaptive mutation.

ADAPTIVEMUTATION produces an offspring s' from a parent s using asymmetric bit-flip rates that are annealed over time and capped by remaining budget. It first sets the baseline mutation rate $p_0 = \alpha/n$ (line 2) and defines a time- and diversity-modulated rate p (lines 3–4). It then sets deletion and addition rates asymmetrically (line 5), since deletions are always budget-feasible whereas additions must be controlled by the remaining slack. To respect the budget, it computes the current cost and slack (lines 6–8) and converts slack into an upper bound q_{slack} on the addition rate using the remaining cost mass (line 9). With small probability

Algorithm 3 POMC_GR

Input: Index i; budget B; horizon $T \cdot G$; objective f; cost c
Output: Best feasible solution x with $c(x) \leq B$
Initialize Pareto set $\mathcal{P} \leftarrow \{\mathbf{0}\}$.
Seed $\mathcal{P}$ with feasible singletons and their greedy extensions (excluding i).
for $t \leftarrow 1$ **to** $T \cdot G$ **do**
  Sample parent $x \in \mathcal{P}$ uniformly.
  $y \leftarrow$ ADAPTIVEMUTATION$(x, t, \mathcal{P})$
  Enforce $y_i \leftarrow 0$.
  if $c(y) = 0$ **or** $c(y) > B$ **then**
    continue
  end if
  if $\exists z \in \mathcal{P} :\ z \succ y$ **then**
    continue
  end if
  $\mathcal{P} \leftarrow \{y\} \cup \{z \in \mathcal{P} :\ \neg(y \succeq z)\}$.
end for
return $\arg\max_{x \in \mathcal{P}} \{f(x) : c(x) \leq B\}$.

p_b, it triggers an exploration burst to escape local minima, and caps p_{add} by q_{slack} so burst-induced additions cannot overwhelm the remaining budget slack (line 10). Finally, it performs asymmetric bitwise mutation: for each coordinate i, it flips s_i with probability p_{del} if $s_i = 1$ and with probability p_{add} if $s_i = 0$, returning the resulting offspring s' (lines 11–18).

2.2. Theoretical Aspects

EPOL is a special case of EPOADAPT, and as such carries its approximate theoretical guarantees.

Proposition 2.4. *Consider* EPOADAPT under the following specialization: (i) all residual problems $(V \setminus \{v\}, f(\cdot \mid v), CS, B - CS(\mathbf{e}_v))$ are enumerated for every $v \in V$; (ii) the anchor set is fixed to $\mathcal{I} = V$ (equivalently $K_B = n$); (iii) each residual problem is initialized as in POMC and uses standard bitwise mutation with flip probability $1/n$; (iv) all additional heuristics are disabled so that the inner solver is exactly POMC.

Under these settings, EPOADAPT reduces exactly to EPOL (Algorithm 2 in (Liu & Qian, 2025)).

Proof. Under the stated specialization, EPOADAPT enumerates the same family of residual problems as EPOL and applies POMC with identical initialization, mutation, and selection rules to each instance. Since all additional heuristics are disabled, the resulting algorithmic behavior coincides exactly with EPOL. □

As a consequence, by Theorem 4.1 of (Liu & Qian, 2025),

Confidential reviewer copy. This manuscript is under double-blind review by ICML 2026. Unauthorized sharing, redistribution, or disclosure is strictly prohibited.

Algorithm 4 ADAPTIVEMUTATION: Asymmetric, Budget-Aware Mutation

Input: n; parent $s \in \{0,1\}^n$; generation t; horizon $T_{\max}$; diversity D; parameters α, β; burst probability p_b; burst multiplier m_b; max rate $p_{\max}$; cost vector $c \in \mathbb{R}_+^n$; budget B; slack fraction $\kappa \in (0,1]$.
Output: offspring $s' \in \{0,1\}^n$.
$p_0 \leftarrow \alpha/n$
$p \leftarrow p_0(1-D)\Big(1+\beta\frac{T_{\max}-t}{T_{\max}}\Big)$
$p \leftarrow \min\{p,\, p_{\max}\}$
$p_{\text{del}} \leftarrow \max\{p,\, p_0\}$; $p_{\text{add}} \leftarrow p$
$\text{currentCost} \leftarrow \sum_{i:s_i=1} c_i$
$\text{slack} \leftarrow B - \text{currentCost}$
$\text{remainingCostMass} \leftarrow \sum_{i:s_i=0} c_i$
Set q_{slack} based on slack and remainingCostMass (0 if remainingCostMass $= 0$, p_0 if slack ≤ 0, else $\kappa \cdot \text{slack}/\text{remainingCostMass}$).
With probability p_b, set p_{del} to $\min\{m_b p_{\text{del}},\, p_{\max}\}$ and p_{add} to $\min\{m_b p_{\text{add}},\, p_{\max},\, q_{\text{slack}}\}$; if $q_{\text{slack}} \ge p_0$ then additionally enforce p_{add} to be at least p_0.
Sample $u_i \sim \text{Unif}(0,1)$ independently as needed.
for $i = 1$ **to** n **do**
  if $s_i = 1$ **then**
    $s'_i \leftarrow s_i \oplus \mathbb{I}[u_i < p_{\text{del}}]$
  else
    $s'_i \leftarrow s_i \oplus \mathbb{I}[u_i < p_{\text{add}}]$
  end if
end for
return s'

the specialized instance of EPOADAPT returns a feasible set $X \subseteq V$ with $CS(X) \le B$ and $f(X) \ge 0.6174\, f(X^\star)$ within the same expected-iteration bound as EPOL.

Lemma 2.5 motivates estimating singleton influence via RIS and using the resulting efficiency scores $\widehat{e}_i := \widehat{\sigma}(\{i\})/c_i$ for each feasible item i.

Lemma 2.5. *Let f be normalized ($f(\emptyset) = 0$), monotone, and submodular, and let c be additive with $c_i > 0$. For any feasible set S with $c(S) \le B$, there exists $j \in S$ such that*

$$\frac{f(\{j\})}{c_j} \;\ge\; \frac{f(S)}{c(S)} \;\ge\; \frac{f(S)}{B}.$$

In particular, for an optimal budget-feasible solution $S^\star$ with $\mathrm{OPT}_B = f(S^\star)$, there exists $j \in S^\star$ such that

$$\frac{f(\{j\})}{c_j} \;\ge\; \frac{\mathrm{OPT}_B}{B}.$$

Proof. Fix an ordering $S = \{i_1, \dots, i_m\}$ and define $S_t := \{i_1, \dots, i_t\}$ with $S_0 = \emptyset$. By submodularity and normalization $f(\emptyset) = 0$, the marginal gain of i_t is largest at the empty set, hence

$$f(S_t) - f(S_{t-1}) \le f(\{i_t\}). \tag{1}$$

Summing (1) over $t = 1, \dots, m$ yields

$$f(S) = \sum_{t=1}^{m}\big(f(S_t) - f(S_{t-1})\big) \le \sum_{t=1}^{m} f(\{i_t\}) = \sum_{i \in S} f(\{i\}).$$

Now let $r := \max_{i \in S} f(\{i\})/c_i$. Then $f(\{i\}) \le r\, c_i$ for all $i \in S$, hence

$$f(S) \le \sum_{i \in S} f(\{i\}) \le r \sum_{i \in S} c_i = r\, c(S).$$

Therefore $r \ge f(S)/c(S) \ge f(S)/B$, proving that some $j \in S$ satisfies $f(\{j\})/c_j \ge f(S)/B$. Applying the same argument to an optimal feasible set $S^\star$ gives $\max_{j \in S^\star} f(\{j\})/c_j \ge \mathrm{OPT}_B/B$. □

We then keep only the top M items under $\widehat{e}_i$ (we use $M = 20$) to cap the downstream compute: both the overlap-penalized greedy anchor construction and the number of parallel anchored SUBPOMC runs scale with the candidate pool size.

We now discuss the feasibility-oriented guarantees for ADAPTIVEMUTATION. We begin with a slack-controlled bound on the expected added cost. The constant $\kappa \in (0,1]$ in the following discussion, controls how aggressively bits may be added under limited slack.

Lemma 2.6. *Let $s \in \{0,1\}^n$ be a feasible solution with slack $\Delta := B - CS(s) > 0$, and let $Z = \{i : s_i = 0\}$. Suppose* ADAPTIVEMUTATION *flips each zero-bit $i \in Z$ independently with probability p_{add}, and define*

$$A \;:=\; \sum_{i \in Z} c_i\, \xi_i, \qquad \xi_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p_{\text{add}}).$$

If $p_{\text{add}} \le \kappa\, \Delta / \sum_{i \in Z} c_i$, then $\mathbb{E}[A] \le \kappa\, \Delta$.

Proof.

$$\mathbb{E}[A] = \sum_{i \in Z} c_i\, \mathbb{E}[\xi_i] = p_{\text{add}} \sum_{i \in Z} c_i \le \kappa\, \Delta.$$

□

Next, we show that the slack cap yields a direct probabilistic guarantee that 0→1 flips are unlikely to exceed the remaining slack.

Theorem 2.7. *Under the conditions of Lemma 2.6,*

$$\mathbb{P}(A \ge \Delta) \le \kappa.$$

Proof. Since $A \ge 0$, Markov's inequality gives $\mathbb{P}(A \ge \Delta) \le \mathbb{E}[A]/\Delta$. Lemma 2.6 implies $\mathbb{E}[A]/\Delta \le \kappa$. □

Confidential reviewer copy. This manuscript is under double-blind review by ICML 2026. Unauthorized sharing, redistribution, or disclosure is strictly prohibited.

In other words, under the slack cap, additions exceed the available slack with probability at most κ. Under a fixed iteration cap, performance is dominated by near-boundary states where the remaining slack Δ is small, many $0 \to 1$ flips select items with $c_i > \Delta$, and standard $1/n$ bit-wise mutation produce budget-violating offspring with non-negligible probability.

Proposition 2.8. *Let $s \in \{0,1\}^n$ be feasible with slack $\Delta := B - CS(s) > 0$, and let $Z = \{i : s_i = 0\}$. Consider standard bitwise mutation with flip probability $1/n$ applied to s, and let*

$$A \;:=\; \sum_{i \in Z} c_i\, \xi_i, \qquad \xi_i \overset{\text{indep.}}{\sim} \text{Bernoulli}(1/n),$$

denote the added cost mass from $0 \to 1$ flips, ignoring deletions. Define the set of individually infeasible additions

$$H \;:=\; \{\, i \in Z : c_i > \Delta \,\}.$$

Then

$$\mathbb{P}(A \geq \Delta) \;\geq\; 1 - (1 - 1/n)^{|H|}.$$

Proof. If any bit in H flips from 0 to 1, then $A \geq c_i > \Delta$, hence $A \geq \Delta$. Therefore $\{A < \Delta\}$ implies that no bit in H flipped. Since each $i \in H$ flips independently with probability $1/n$, the probability that none flips is $(1 - 1/n)^{|H|}$, which yields $\mathbb{P}(A \geq \Delta) \geq 1 - (1 - 1/n)^{|H|}$. □

This bound has an immediate fixed-cap consequence, since $A \geq \Delta$ already implies $CS(s') > B$, and a constant-sized infeasible region among the zero-bits forces a constant rejection rate.

Corollary 2.9. *In the setting of Proposition 2.8, let $S = \{i : s_i = 1\}$ and $m := |S|$. Assume $|H| \geq \gamma n$ and $m \leq \beta n$ for constants $\gamma > 0$ and $\beta \in (0, 1]$. Then for all $n \geq 2$,*

$$\mathbb{P}(CS(s') > B) \;\geq\; (1 - e^{-\gamma})\, e^{-2\beta}.$$

Proof. Let E be the event that at least one bit in H flips from 0 to 1, and let F be the event that no bit in S flips from 1 to 0. If $E \cap F$ occurs, then some $i \in H$ is added, so the added cost satisfies $A \geq c_i > \Delta$, and since F implies no deletions, $CS(s') = CS(s) + A > B$. Thus,

$$\mathbb{P}(CS(s') > B) \;\geq\; \mathbb{P}(E \cap F).$$

Events E and F depend on disjoint coordinates, hence they are independent under bitwise mutation, so

$$\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F) = \Big(1 - (1 - 1/n)^{|H|}\Big)\,(1 - 1/n)^m.$$

Using $|H| \geq \gamma n$ gives $(1 - 1/n)^{|H|} \leq (1 - 1/n)^{\gamma n} \leq e^{-\gamma}$, hence

$$1 - (1 - 1/n)^{|H|} \geq 1 - e^{-\gamma}.$$

Also $\ln(1 - x) \geq -x/(1 - x)$ for $x \in (0, 1)$, so with $x = 1/n$,

$$(1 - 1/n)^m = \exp\big(m \ln(1 - 1/n)\big) \geq \exp\left(-\frac{m}{n - 1}\right).$$

If $m \leq \beta n$ and $n \geq 2$, then $\frac{n}{n-1} \leq 2$, so

$$(1 - 1/n)^m \geq \exp\left(-\frac{\beta n}{n - 1}\right) \geq e^{-2\beta}.$$

Multiplying the two bounds yields $\mathbb{P}(CS(s') > B) \geq (1 - e^{-\gamma})e^{-2\beta}$. □

Corollary 2.9 formalizes a fixed-cap inefficiency of standard $1/n$ mutation near the budget boundary. When a linear fraction of zero-bits are individually infeasible and the current solution size is $O(n)$, each mutation step is rejected with constant probability bounded below by $(1 - e^{-\gamma})e^{-2\beta}$, so a constant fraction of a finite iteration budget is spent generating offspring that are discarded before any objective evaluation.

We use RIS for large-scale screening and reserve a small Monte Carlo (MC) budget only for a final refinement step. The choice $R = 50$ follows from the standard $1/\sqrt{R}$ concentration of bounded IC outcomes, yielding diminishing returns beyond moderate R under a fixed runtime cap.

Lemma 2.10. *Fix a seed set $S \subseteq V$. Let $Z \in \{0, 1, \ldots, n\}$ be the number of activated nodes in one IC simulation initialized at S, and let $Z_1, \ldots, Z_R$ be i.i.d. copies. Define the normalized random variable $Y_r := Z_r/n \in [0, 1]$, its mean $f(S) := \mathbb{E}[Y_1]$, and the MC estimator*

$$\widehat{f}_R(S) := \frac{1}{R} \sum_{r=1}^{R} Y_r.$$

Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\big|\widehat{f}_R(S) - f(S)\big| \;\leq\; \sqrt{\frac{\ln(2/\delta)}{2R}}.$$

Proof. Since $Y_1, \ldots, Y_R$ are i.i.d. and bounded in $[0, 1]$, Hoeffding's inequality gives, for any $t > 0$,

$$\mathbb{P}\Big(\Big|\widehat{f}_R(S) - f(S)\Big| \geq t\Big) \leq 2e^{-2Rt^2}.$$

Choosing $t = \sqrt{\ln(2/\delta)/(2R)}$ so that $2e^{-2Rt^2} = \delta$ and substituting into the bound proves the claim. □

For $\delta = 0.05$, the bound becomes

$$\big|\widehat{f}_R(S) - f(S)\big| \;\leq\; \sqrt{\frac{\ln(40)}{2R}}.$$

Confidential reviewer copy. This manuscript is under double-blind review by ICML 2026. Unauthorized sharing, redistribution, or disclosure is strictly prohibited.

For reference, this yields the following worst-case error bars:

$$R = 10 \quad : \quad \varepsilon \leq 0.43,$$
$$R = 50 \quad : \quad \varepsilon \leq 0.19,$$
$$R = 100 \quad : \quad \varepsilon \leq 0.14.$$

Thus moving from $R = 10$ to $R = 50$ tightens the bound by $\sqrt{5}$, while moving from $R = 50$ to $R = 100$ tightens it only by $\sqrt{2}$ at twice the refinement cost; we therefore use $R = 50$ as a practical elbow.

3. Experiments

We evaluate on three real-world graph datasets commonly used in IM, namely GRAPH100 (100 vertices, 3,465 edges), GRAPH200 (200 vertices, 9,950 edges), and the animal interaction network INSECTA (152 vertices, 6,716 edges). We use exactly the same datasets as EPOL (Liu & Qian, 2025), enabling direct table-level comparison for IM. Following EPOL, we consider budget values $B \in \{100, 200, 300, 400, 500\}$. Most experiments were conducted on Google Colab Pro using an AMD EPYC 7B12 CPU, while the largest-budget runs at edge probability 0.1 utilized AMD EPYC 7702 CPUs hosted at a high-performance computing center. The CPU affects only wall-clock time, whereas solution quality is governed by the evaluation limit: for a fixed limit such as $4nK_B$ or $20nK_B$, the algorithm executes the same number of evaluations independent of hardware, and faster CPUs simply reach the limit sooner. Unlike EPOL, which allocates a runtime budget of $20nK_B$, EPOADAPT is evaluated under a strictly tighter budget of $4nK_B$, corresponding to a fivefold reduction in allowed evaluations across all experiments; despite this stringent evaluation limit, EPOADAPT consistently achieves strictly higher influence values than those reported by EPOL across all budgets and for both edge probabilities considered. Moreover, all reported results are averaged over at least three independent runs, with additional runs performed as needed until the coefficient of variation fell well below 10%, ensuring that the reported performance differences are statistically stable.

[1]

3.1. Edge probability 0.05

We begin by evaluating the performance of EPOADAPT against the evolutionary algorithms that scale as $\mathcal{O}(B)$, namely POMC (Qian et al., 2017), FPOMC (Bian et al., 2021), and EPOL (Liu & Qian, 2025), using a fixed IM budget $B = 100$. This comparison is conducted on identical hardware to provide a controlled environment for runtime analysis. Figure 1 reports both the objective value and the wall-clock runtime (log scale) across all three datasets.

[1]See https://github.com/icml2026/EpoAdapt for code and reproducibility.

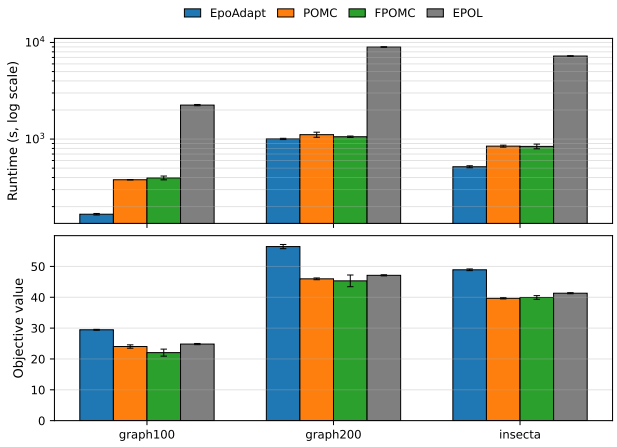

Figure 1. $p = 0.05$, $B = 100$: EPOADAPT vs. POMC(Qian et al., 2017), FPOMC(Bian et al., 2021), EPOL(Liu & Qian, 2025) on GRAPH100, GRAPH200, and INSECTA. Runtime is shown in log-scale (top) and objective value (bottom), with identical hardware across methods and a $4nK_B$ computation limit.

Figure 1 illustrates the strict dominance of EPOADAPT at $B = 100$: it achieves the highest objective value and the lowest wall-clock runtime across all three datasets among the compared methods. Most notably on GRAPH200, EPOADAPT surpasses EPOL(Liu & Qian, 2025) in objective value while reducing runtime by approximately an order of magnitude.

Table 1. Influence Maximization for edge probability of 0.05 and budgets $\{100, 200, \cdots, 500\}$. Best results are bolded.

	100	200	300	400	500
graph100 (100 vertices, 3,465 edges)					
GGA	22.43 ± 1.21	35.49 ± 1.35	43.30 ± 0.99	49.71 ± 1.10	55.18 ± 0.89
Greedy+	23.87 ± 0.37	35.61 ± 0.78	44.22 ± 0.63	50.92 ± 0.61	55.80 ± 0.69
1-guess	24.66 ± 0.17	37.17 ± 0.29	45.70 ± 0.33	51.89 ± 0.18	56.91 ± 0.26
EVO-SMC	25.03 ± 0.32	37.39 ± 0.55	44.72 ± 0.79	50.45 ± 0.92	54.44 ± 1.02
FPOMC	24.29 ± 0.64	36.58 ± 1.12	45.18 ± 0.95	51.24 ± 0.70	55.74 ± 0.88
EAMC	25.08 ± 0.27	37.80 ± 0.18	45.88 ± 0.44	51.19 ± 0.56	56.21 ± 0.34
POMC	24.96 ± 0.35	37.56 ± 0.39	45.81 ± 0.30	52.24 ± 0.21	57.03 ± 0.22
EPOL	25.48 ± 0.17	37.98 ± 0.17	46.41 ± 0.24	52.76 ± 0.14	57.71 ± 0.17
EpoAdapt	**29.46 ± 0.14**	**41.37 ± 0.42**	**49.05 ± 0.26**	**54.39 ± 0.36**	**59.33 ± 0.56**
graph200 (200 vertices, 9,950 edges)					
GGA	44.31 ± 1.14	70.87 ± 2.84	89.51 ± 2.55	103.44 ± 2.64	109.07 ± 3.16
Greedy+	45.12 ± 0.47	81.57 ± 0.72	94.06 ± 0.91	106.95 ± 1.01	115.22 ± 1.88
1-guess	46.11 ± 0.52	82.56 ± 0.75	96.58 ± 0.69	108.97 ± 0.33	117.47 ± 0.28
EVO-SMC	48.04 ± 0.44	84.38 ± 0.21	96.81 ± 0.60	109.09 ± 1.66	116.77 ± 1.78
FPOMC	46.72 ± 0.77	82.62 ± 2.57	95.92 ± 0.67	108.54 ± 0.90	117.64 ± 0.91
EAMC	47.92 ± 0.43	84.78 ± 0.44	97.06 ± 0.46	110.81 ± 0.27	118.37 ± 2.06
POMC	46.78 ± 0.61	83.87 ± 0.66	98.41 ± 0.59	110.05 ± 0.73	119.40 ± 0.24
EPOL	47.79 ± 0.33	84.76 ± 0.35	99.73 ± 0.26	111.04 ± 0.43	120.12 ± 0.26
EpoAdapt	**56.44 ± 0.68**	**93.60 ± 0.65**	**107.85 ± 0.53**	**117.53 ± 0.78**	**125.32 ± 0.73**
insecta (152 vertices, 6,716 edges)					
GGA	38.09 ± 1.33	53.35 ± 4.24	65.48 ± 3.31	75.55 ± 1.39	83.17 ± 1.49
Greedy+	39.49 ± 1.07	57.94 ± 0.52	68.86 ± 2.65	78.39 ± 1.31	84.69 ± 1.04
1-guess	40.57 ± 0.64	58.99 ± 0.27	72.90 ± 0.66	80.24 ± 0.39	86.59 ± 0.26
EVO-SMC	41.76 ± 0.28	59.63 ± 0.29	68.73 ± 0.87	78.14 ± 1.29	85.11 ± 0.62
FPOMC	40.48 ± 0.57	58.75 ± 0.39	71.70 ± 1.82	79.75 ± 0.41	85.92 ± 0.89
EAMC	41.79 ± 0.18	59.66 ± 0.35	70.81 ± 1.18	80.04 ± 0.68	86.48 ± 0.41
POMC	40.90 ± 0.51	58.40 ± 0.57	73.96 ± 0.40	81.10 ± 0.42	87.15 ± 0.28
EPOL	41.64 ± 0.26	59.89 ± 0.33	74.60 ± 0.20	81.77 ± 0.15	88.00 ± 0.07
EpoAdapt	**48.91 ± 0.27**	**65.91 ± 0.69**	**79.83 ± 0.40**	**85.60 ± 0.36**	**91.57 ± 0.64**

Table 1 reports IM results for $p = 0.05$ over budgets

Confidential reviewer copy. This manuscript is under double-blind review by ICML 2026. Unauthorized sharing, redistribution, or disclosure is strictly prohibited.

$B \in \{100, 200, 300, 400, 500\}$. With the exception of EPOADAPT, all entries are taken directly from EPOL (Liu & Qian, 2025) to enable direct table-level comparison under the same datasets and evaluation protocol. Table 1 shows that EPOADAPT consistently outperforms all nine compared methods across every tested instance. The dominance is particularly evident in larger instances; for example, on GRAPH200 with $B = 500$, EPOADAPT achieves an objective value of 125.32 ± 0.73, compared to 120.12 ± 0.28 for EPOL(Liu & Qian, 2025). These results confirm that our adaptive approach consistently identifies higher-quality seed sets under a substantially tighter evaluation limit.

3.2. Edge probability 0.1

We next consider edge probability $p = 0.1$ under the same IC model and experimental protocol. Relative to $p = 0.05$, this setting induces denser cascade realizations, so it provides a complementary regime in which each objective evaluation reflects larger expected propagation. We retain the same budgets and evaluation limits as in the $p = 0.05$ experiments.

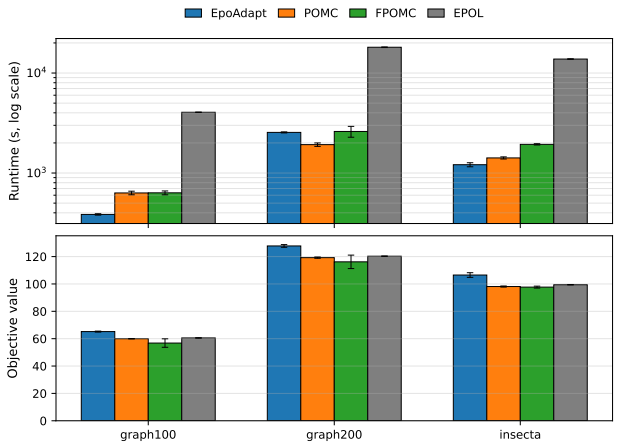

Figure 2. $p = 0.1$, $B = 100$: EPOADAPT vs. POMC(Qian et al., 2017), FPOMC(Bian et al., 2021), EPOL(Liu & Qian, 2025) on GRAPH100, GRAPH200, and INSECTA. Runtime is shown in log-scale (top) and objective value (bottom), with identical hardware across methods and a $4nK_B$ computation limit.

At $p = 0.1$ and $B = 100$, Figure 2 shows that EPOADAPT simultaneously attains the highest objective value and the lowest wall-clock runtime across all three datasets. The runtime separation is most pronounced on GRAPH200, where EPOADAPT is about an order of magnitude faster than EPOL (Liu & Qian, 2025). Table 2 summarizes objective values over $B \in \{100, 200, 300, 400, 500\}$. As with table 1, with the exception of EPOADAPT, all entries are taken directly from EPOL (Liu & Qian, 2025) to enable direct table-level comparison under the same datasets and evaluation protocol. We see from Table 2 that EPOADAPT attains the best value in all 15 settings; on GRAPH200 with $B = 500$, it improves over EPOL (Liu & Qian, 2025) from 174.99 ± 0.09 to 177.74 ± 0.68.

Table 2. Influence Maximization for edge probability of 0.1 and budgets $\{100, 200, \cdots, 500\}$. Best results are bolded.

	100	200	300	400	500
graph100 (100 vertices, 3,465 edges)					
GGA	58.49 ± 0.50	66.18 ± 1.78	74.17 ± 0.87	78.58 ± 1.03	81.76 ± 0.88
Greedy+	58.28 ± 0.38	68.71 ± 0.97	75.42 ± 0.63	79.47 ± 0.58	82.62 ± 0.42
1-guess	59.66 ± 0.26	71.49 ± 0.25	76.79 ± 0.22	80.33 ± 0.17	83.36 ± 0.19
EVO-SMC	60.22 ± 0.41	70.34 ± 1.26	75.29 ± 1.01	79.81 ± 0.28	82.53 ± 0.56
FPOMC	59.05 ± 0.90	70.44 ± 1.16	76.40 ± 0.43	80.20 ± 0.62	83.07 ± 0.43
EAMC	60.40 ± 0.35	71.02 ± 0.74	75.93 ± 0.90	80.07 ± 0.41	83.79 ± 0.37
POMC	59.97 ± 0.16	71.77 ± 0.16	76.92 ± 0.12	80.83 ± 0.27	83.80 ± 0.16
EPOL	60.65 ± 0.18	72.13 ± 0.12	77.28 ± 0.06	81.37 ± 0.09	84.09 ± 0.05
EpoAdapt	$\mathbf{65.19 \pm 0.34}$	$\mathbf{74.83 \pm 0.57}$	$\mathbf{79.14 \pm 0.26}$	$\mathbf{82.24 \pm 0.34}$	$\mathbf{85.01 \pm 0.20}$
graph200 (200 vertices, 9,950 edges)					
GGA	115.79 ± 0.93	148.37 ± 3.00	159.45 ± 1.22	166.47 ± 2.82	168.26 ± 1.58
Greedy+	116.14 ± 1.35	153.05 ± 0.76	162.60 ± 0.64	169.17 ± 0.88	172.00 ± 1.10
1-guess	118.53 ± 0.47	154.32 ± 0.57	163.57 ± 0.17	170.43 ± 0.21	173.87 ± 0.45
EVO-SMC	120.36 ± 0.30	155.23 ± 0.12	163.78 ± 0.22	169.63 ± 0.89	172.37 ± 1.35
FPOMC	119.88 ± 0.35	155.26 ± 0.17	163.82 ± 0.26	170.70 ± 0.35	174.39 ± 0.57
EAMC	120.66 ± 0.45	155.32 ± 0.18	163.93 ± 0.22	170.79 ± 0.45	174.36 ± 0.75
POMC	119.65 ± 0.25	154.91 ± 0.20	163.54 ± 0.18	170.70 ± 0.35	174.72 ± 0.07
EPOL	120.61 ± 0.19	155.55 ± 0.28	164.15 ± 0.29	171.14 ± 0.09	174.99 ± 0.09
EpoAdapt	$\mathbf{127.78 \pm 0.98}$	$\mathbf{160.46 \pm 0.59}$	$\mathbf{168.61 \pm 0.43}$	$\mathbf{174.04 \pm 0.37}$	$\mathbf{177.74 \pm 0.68}$
insecta (152 vertices, 6,716 edges)					
GGA	95.34 ± 1.07	107.39 ± 2.65	115.54 ± 1.53	122.04 ± 1.04	126.28 ± 1.00
Greedy+	95.85 ± 0.83	111.41 ± 0.96	120.15 ± 1.75	124.18 ± 0.44	128.16 ± 0.40
1-guess	97.94 ± 0.34	113.09 ± 0.38	121.65 ± 0.25	125.13 ± 0.31	129.05 ± 0.18
EVO-SMC	99.37 ± 0.36	113.54 ± 0.56	119.45 ± 0.74	124.55 ± 1.08	128.02 ± 0.76
FPOMC	98.83 ± 0.29	112.95 ± 0.52	122.11 ± 0.13	125.50 ± 0.47	129.33 ± 0.27
EAMC	99.28 ± 0.31	113.93 ± 0.18	120.58 ± 1.01	126.21 ± 0.53	128.80 ± 0.42
POMC	98.84 ± 0.31	113.86 ± 0.33	122.06 ± 0.21	126.43 ± 0.20	129.52 ± 0.13
EPOL	99.52 ± 0.18	114.25 ± 0.15	122.38 ± 0.11	126.69 ± 0.08	129.71 ± 0.09
EpoAdapt	$\mathbf{106.52 \pm 1.69}$	$\mathbf{116.85 \pm 0.59}$	$\mathbf{124.38 \pm 0.37}$	$\mathbf{128.20 \pm 0.29}$	$\mathbf{130.95 \pm 0.08}$

4. Conclusion

We introduced EPOADAPT, an adaptive evolutionary Pareto optimizer for monotone submodular maximization under linear cost constraints, and demonstrated its effectiveness on the budgeted IM problem under the IC model. The key insight driving EPOADAPT is that in practice, computational resources are finite, and the central question becomes whether we can find high-quality solutions within these limits. Traditional approaches like EPOL rely on uniform exploration and pure Monte Carlo objective estimation. In contrast, EPOADAPT employs RIS-based screening under theoretical guarantees to identify promising regions, refines objective estimates through lightweight Monte Carlo sampling, and adapts the anchor count K_B in a submodularity-guided way to manage the exploration–exploitation tradeoff across parallel subproblems. Feasibility near the constraint boundary is promoted through statistically grounded asymmetric mutation. Since EPOL emerges as a special case of our framework, the existing approximation guarantees carry over under the corresponding parameter settings. Empirically, on standard benchmarks evaluated under stricter computational limits, EPOADAPT consistently achieves superior objective values, demonstrating that submodularity-guided adaptation of K_B and statistically grounded feasibility promotion during variation yield more effective use of finite computational resources.

Confidential reviewer copy. This manuscript is under double-blind review by ICML 2026. Unauthorized sharing, redistribution, or disclosure is strictly prohibited.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

References

Ashkan, A., Kveton, B., Berkovsky, S., and Wen, Z. Optimal greedy diversity for recommendation. In *IJCAI*, volume 15, pp. 1742–1748, 2015.

Badanidiyuru, A. and Vondrák, J. Fast algorithms for maximizing submodular functions. In *Proceedings of the twenty-fifth annual ACM-SIAM symposium on Discrete algorithms*, pp. 1497–1514. SIAM, 2014.

Bian, C., Feng, C., Qian, C., and Yu, Y. An efficient evolutionary algorithm for subset selection with general cost constraints. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, 2020.

Bian, C., Qian, C., and Tang, K. Fast pareto optimization for subset selection with dynamic cost constraints. In *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2021.

Borgs, C., Brautbar, M., Chayes, J., and Lucier, B. Maximizing social influence in nearly optimal time. In *Proceedings of the twenty-fifth annual ACM-SIAM symposium on Discrete algorithms*. Society for Industrial and Applied Mathematics, 2014.

Ene, A. and Nguyen, H. L. A nearly-linear time algorithm for submodular maximization with a knapsack constraint. In *Proceedings of the 46th International Colloquium on Automata, Languages, and Programming (ICALP 2019)*, pp. 53:1–53:12, Patras, Greece, 2019.

Feige, U. A threshold of ln n for approximating set cover. *Journal of the ACM (JACM)*, 45(4):634–652, 1998.

Feldman, M., Nutov, Z., and Shoham, E. Practical budgeted submodular maximization. *Algorithmica*, 85(5):1332–1371, 2023.

Kempe, D., Kleinberg, J., and Tardos, É. Maximizing the spread of influence through a social network. In *Proceedings of the ninth ACM SIGKDD international conference on Knowledge discovery and data mining*, 2003.

Krause, A. and Guestrin, C. A note on the budgeted maximization of submodular functions. Technical report, Carnegie Mellon University, Center for Automated Learning and Discovery, 2005.

Krause, A., Leskovec, J., Guestrin, C., VanBriesen, J., and Faloutsos, C. Efficient sensor placement optimization for securing large water distribution networks. *Journal of Water Resources Planning and Management*, 134(6): 516–526, 2008.

Lin, H. and Bilmes, J. Multi-document summarization via budgeted maximization of submodular functions. In *Human Language Technologies: The 2010 Annual Conference of the North American Chapter of the Association for Computational Linguistics*, pp. 912–920, 2010.

Lin, H. and Bilmes, J. A class of submodular functions for document summarization. In *Proceedings of the 49th annual meeting of the association for computational linguistics: human language technologies*, pp. 510–520, 2011.

Liu, D.-X. and Qian, C. Improved theoretically-grounded evolutionary algorithms for subset selection with a linear cost constraint. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025.

Liu, D.-X., Mu, X., and Qian, C. Human assisted learning by evolutionary multi-objective optimization. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, pp. 12453–12461, 2023. Vol. 37, No. 10.

Qian, C., Shi, J.-C., Yu, Y., and Tang, K. On subset selection with general cost constraints. In *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, volume 17, 2017.

Sviridenko, M. A note on maximizing a submodular set function subject to a knapsack constraint. *Operations Research Letters*, 32(1):41–43, 2004.

Tang, Y., Xiao, X., and Shi, Y. Influence maximization: Near-optimal time complexity meets practical efficiency. In *Proceedings of the 2014 ACM SIGMOD international conference on Management of data*, pp. 75–86, 2014.

Tang, Y., Shi, Y., and Xiao, X. Influence maximization in near-linear time: A martingale approach. In *Proceedings of the 2015 ACM SIGMOD international conference on management of data*, 2015.

Tropp, J. A. Greed is good: Algorithmic results for sparse approximation. *IEEE Transactions on Information theory*, 50(10):2231–2242, 2004.

Yaroslavtsev, G., Zhou, S., and Avdiukhin, D. "bring your own greedy"+ max: near-optimal 1/2-approximations for submodular knapsack. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*. PMLR, 2020.

Confidential reviewer copy. This manuscript is under double-blind review by ICML 2026. Unauthorized sharing, redistribution, or disclosure is strictly prohibited.

Zhang, H. and Vorobeychik, Y. Submodular optimization with routing constraints. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, pp. 3084–3090, 2016. Vol. 30, No. 1.

Zhu, Y., Basu, S., and Pavan, A. Improved evolutionary algorithms for submodular maximization with cost constraints. *arXiv preprint arXiv:2405.05942*, 2024.

Confidential reviewer copy. This manuscript is under double-blind review by ICML 2026. Unauthorized sharing, redistribution, or disclosure is strictly prohibited.